

Executive Summary

The UK currently imports virtually all its titanium metal and alloys, leaving critical aerospace, defence, and energy applications exposed to supply disruptions (EU imports ~60% of titanium wrought products [26†L9-L17] [26†L27-L30]). Building a homegrown production capability would improve security of supply and decarbonise production (via UK renewables/hydrogen). Sheffield is well-placed to host such industry: it has a world-class advanced manufacturing cluster (AMRC, Boeing, Rolls-Royce, etc. nearby [23†L12-L23]) and a skilled metallurgical workforce from its steel heritage. This proposal outlines a phased development path (from R&D through pilot, demonstration, and full-scale) to make Sheffield a major titanium sponge and alloy producer for UK markets. We review production technologies (Kroll, electrochemical, novel low-carbon routes), estimate costs and inputs, map supply chains, and analyse UK demand by sector. Key enablers include leveraging Sheffield's innovation ecosystem (e.g. Metalysis' FFC technology [39†L132-L136]), securing funding and regulatory support, and phased pilot projects. Risks (technical scale-up, market competition, permitting) can be mitigated by collaboration with industry bodies (ADS, UKRI) and clear policy incentives.

Objectives

- **Establish Titanium Production** – Evaluate and plan how Sheffield can produce titanium metal (sponge or powder) and alloys at scale for the UK.
- **Technology Selection** – Survey production methods (Kroll, Hunter, electrochemical FFC, hydrometallurgical, recycling, etc.) and choose or develop processes suitable for UK scale-up (with low carbon-intensity).
- **Cost & Resources** – Estimate capital and operating costs, energy use, and raw-material requirements for proposed processes.
- **Regulatory Compliance** – Identify UK/EU permitting, environmental, and safety regulations (e.g. EPR permits, COMAH, UKCA certification) relevant to a titanium plant.
- **Workforce & Skills** – Assess Sheffield's existing skills base and training needs for titanium metallurgy and chemical processing.
- **Supply Chain Mapping** – Map upstream inputs (titanium ore, magnesium, aluminium, vanadium, chlorine, inert gases, etc.) and logistics (sites, transport, ports) in the Sheffield region.
- **Market Demand & Competitiveness** – Analyse UK titanium demand by sector (aerospace, defence, automotive, medical, energy, additive manufacturing) and pricing vs imports.
- **R&D/IP & Finance** – Determine intellectual property (existing and needed) and R&D gaps, and outline financing models (public/private) and policy incentives (grants, tax credits, carbon pricing).
- **Scale-Up Roadmap** – Develop a phased timeline from pilot to commercial production, with milestones, scale factors (capacity in tonnes/year), and risk mitigation strategies.
- **Recommendations** – Propose next steps, partnerships, and policy actions to initiate the programme.

Background: Titanium Production Technologies

Titanium is extracted from ore (rutile/ilmenite) almost exclusively via processes that convert TiO_2 into metal. The **Kroll process** is the established industrial route [11†L878-L884] [37†L53-L62] : TiO_2 is

chlorinated (with carbon) to TiCl_4 , then purified and reduced by molten magnesium at $\sim 800\text{--}1000^\circ\text{C}$ in argon, yielding porous “Ti sponge” and MgCl_2 waste [11†L878-L884] [37†L53-L62]. Kroll-produced sponge (typically 99.5–99.7% Ti) is then crushed and melted (e.g. vacuum-arc-melted) into ingots or alloy billets. Advantages: produces very pure Ti suitable for aerospace. Disadvantages: batch (not continuous), slow (days per batch), energy-intensive and costly [33†L49-L58] [13†L932-L937]; it still dominates global supply.

An older **Hunter process** (USA, 1950s) used $\text{TiCl}_4 + \text{Na}$ to make Ti, but was more expensive than Kroll [11†L878-L884] and has been superseded. A powder-metallurgy method (Armstrong process) flows $\text{TiCl}_4 + \text{molten Na}$ continuously to make Ti powder [53†L910-L919], but is niche. Other early concepts (e.g. carbothermic reduction of TiO_2) generally fail because TiC forms.

Electrochemical routes seek to skip chlorination. The FFC-Cambridge process (held by Sheffield-based Metalysis [39†L132-L136]) electrolyzes solid TiO_2 in molten CaCl_2 ($900\text{--}1100^\circ\text{C}$) using a carbon anode [39†L117-L125]. Oxygen is removed electrochemically, leaving solid Ti (often as a coherent piece or powder). The process has been demonstrated at lab/pilot scale but not yet commercial; Sheffield’s Metalysis is building pilot plants to scale it up. Other molten-salt electrolysis methods (molten oxide electrolysis, plasma electrolytic) are in R&D [53†L928-L937]. A key benefit of electrochemical routes is avoiding toxic TiCl_4 and potentially using electricity (and an inert anode) to reduce CO_2 footprint.

Hydrogen-assisted magnesiothermic (Hycam) is another novel process: TiO_2 is reduced by Mg under a hydrogen atmosphere, then acid-leached to remove Mg and residues, yielding titanium hydride/powder [13†L942-L950]. This is still at R&D stage but avoids producing TiCl_4 altogether.

Recycling and powder metallurgy: Currently, scrap titanium (from off-cuts or end-of-life) is remelted (VAR or EB furnace) or gas-atomized. A US plant reprocesses Ti scrap into powder (~ 125 t/yr target) [13†L937-L940]. Recycling is limited by existing small volumes of scrap; a UK plant could ramp this up as part of operations. Powder metallurgy (e.g. direct $\text{TiO}_2 \rightarrow \text{powder}$ via modified Armstrong) is less mature.

Table 1 compares key production routes:

Process	Feedstock/ Inputs	Product	Status (Commercial)	Pros/Cons
Kroll (magnesiothermic)	TiO_2 (rutile/ ilmenite), carbon, Cl_2 , Mg metal, Ar	Ti sponge (batch), $\sim 99.5\%+$ Ti	Commercial (global standard)	High purity; well-proven. Cons: energy-intensive (tens kWh/kg), batch process (low throughput), CO_2 (from C), slow/costly [11†L878-L884] [33†L49-L58].
Hunter (sodium)	TiCl_4 , Na metal	Ti sponge (batch)	Obsolete (superseded)	Historically first process; now outcompeted (more expensive than Kroll) [11†L878-L884].

Process	Feedstock/ Inputs	Product	Status (Commercial)	Pros/Cons
Armstrong (fluid Na)	TiCl ₄ gas, molten Na	Ti powder (continuous)	Niche (specialty)	Continuous powder; recovers Na/Cl. Rarely used commercially.
FFC-Cambridge (electrolytic)	TiO ₂ (pellet/cathode), molten CaCl ₂ salt, C anode, ~900–1100 °C	Solid Ti (often powder/porous)	Pilot (Metalysis, UK)	Avoids TiCl ₄ ; potential for lower carbon if powered by renewables. Cons: unproven at scale; reactor/electrolyte materials.
Molten Oxide Electrolysis	TiO ₂ , molten oxide (similar to Hall–Héroult concept)	Ti metal	R&D	Concept (like aluminum process); under research.
Hydrogen-assisted Mg	TiO ₂ powder, Mg metal, H ₂ gas, HCl acid leach	Ti powder or hydride	R&D	Avoids Cl ₂ step; yields powder; still experimental 【13†L942-L950】 .
Electron Beam Remelt (EBM)	Ti sponge/ingot, scrap	High-grade Ti ingots	Commercial (refining stage)	Final alloy refining; improves purity.
Casting/Forging	Melted Ti billets or ingots	Cast parts, forgings	Commercial	Shaping for near-net shapes; less costly but grain structure issues.

Energy and emissions are highest for the Kroll route: estimates range ~50–60 kWh/kg Ti (depending on efficiency) plus thermal energy **【33†L49-L58】** **【13†L932-L937】** . By contrast, electrochemical methods could lower energy per kg (though actual values at scale are uncertain). All routes need inert atmosphere (Ar/He) for melting.

Sheffield’s focus will be on low-carbon methods: e.g. Kroll + carbon capture/renewable energy, or scaling the FFC process with UK renewables. At present **no commercial process fully replaces Kroll** **【53†L928-L937】** , so a prudent approach combines incremental improvement of Kroll with pilot testing of novel routes.

Capital & Operating Costs

A new titanium plant is capital-intensive due to specialized reactors (chlorination, vacuum distillation, Kroll reactors), melting furnaces, and safety systems. For scale, assume initial pilot capacity ~100–500 t/yr, demonstration ~1,000–2,000 t/yr, and large plant 5,000–10,000 t/yr. Rough order-of-magnitude capex (very approximate): a small Kroll-based pilot might cost £50–100M (USD ~\$60–120M), while a full-scale plant could be >£500M. Operating costs break down into raw materials (TiO₂ ore, Mg, etc.), energy, labor, and maintenance. A rough illustrative split for Kroll is: **Magnesium** (~20–30% of op-ex), **Titania feed** (~15%), **Electricity/Heat** (~25–35%), **Labor/maintenance** (~20%), **Argon/consumables** (~5–10%).

Since titanium sells at several times the price of steel, even small cost improvements can be valuable **【33†L49-L58】** .

As a reference, industry commentary notes titanium metal costs an order of magnitude more than steel due to Kroll's complexity **【33†L49-L58】** . If novel processes (electrolytic or hydrogen) materialize, energy (electricity/hydrogen) could dominate costs but at potentially lower carbon. Recovery of byproducts (MgCl_2) and recycling scrap can also offset costs. Ultimately, local production may carry a cost premium over the cheapest imports (e.g. from China/Japan) but offers supply security and lower embodied carbon.

Inputs & Resources

Feedstocks: Primary inputs are titanium minerals (rutile/ilmenite), and alloying elements. Sheffield's plant would **import rutile/ilmenite** by sea: world production is dominated by China (35%), Mozambique (20%), and South Africa (15%) **【11†L783-L792】** . TiO_2 pigment byproduct streams (from ilmenite processing) could be a supplementary source. For alloying, common elements (Al, V, Fe, Mo, etc.) would follow existing supply chains (e.g. Aluminium (UK-based smelters or imports), Vanadium (typ. Australia/China ores), Iron (steel scrap), etc.). **Reducing agents:** Magnesium metal (critical for Kroll) is imported (Norway, China, USA are major Mg producers). Argon gas (for smelting) is available from UK industrial gas suppliers (e.g. Air Products, BOC). Chlorine gas (for TiCl_4 production) could be sourced from domestic chlor-alkali plants.

Energy: Kroll route needs large thermal and electrical inputs. A modern Kroll plant might use natural gas/electric heating for chlorination and Mg reduction (~1–2 MWh thermal and ~0.5 MWh electrical per tonne of Ti). Electrolytic methods use electricity (potentially <5 MWh/t if efficient). For decarbonisation, sourcing low-carbon electricity (grid renewables or dedicated wind/solar) and using hydrogen (for cooling/heat) should be planned. Water and cooling systems must also be sized (for condensers, scrubbers, etc.).

Consumables & Waste: Key consumables include magnesium (to be recycled from MgCl_2), graphite (for electrodes), and acids (for leaching, if used). Kroll produces MgCl_2 as waste, which can be electrolyzed back to Mg (though that requires extra energy). Environmental controls (scrubbers) will capture volatile TiCl_4 , HCl, and recycle or safely dispose of them.

A **supply-chain table** (Table 2) outlines key inputs and likely suppliers:

Input	Source	UK/Sheffield Actors
Titanium ore (rutile/ilmenite)	South Africa, Australia, Canada, China, Mozambique 【11†L783-L792】	Bulk shippers, mineral traders (e.g. Kenmare, Base Resources) handle imports via Humber/Immingham ports.
Titanium dioxide (synthetic rutile)	Processed locally if ore upgraded via Becher (negligible UK output)	Possible tie-ins with pigment plants (e.g. Kronos, Tronox).
Magnesium metal	China, Norway, USA	Chemical distributors (e.g. Pechiney, Eurotitan).

Input	Source	UK/Sheffield Actors
Aluminum	Domestic smelters (Anglesey Aluminium), imports	Metal suppliers (e.g. Rio Tinto Alcan).
Vanadium, Molybdenum	Imports (Australia, China, USA)	Alloy providers (H.C. Starck, Global Trading).
Chlorine (Cl ₂)	UK chlor-alkali plants (e.g. Ineos, Tata)	Bulk chemical distributors.
Argon (Ar), Helium	Industrial gas suppliers (Air Liquide, Linde UK)	Local cylinder filling/plants.
Electricity	UK grid (renewables, nuclear, gas mix)	National Grid (backup); potential on-site renewables.
Gas/Heat (NG or H ₂)	National grid; future green hydrogen	Utility companies; hydrogen suppliers (in future).
Catalyst/ Equipment	Refractory linings, heat exchangers	UK engineering firms (special furnaces).

Environmental, Emissions & Permitting

A titanium plant has several environmental aspects: high energy use, chemical emissions, and solid waste. **Emissions** include CO₂ (from burning fossil fuel or carbon in chlorination/Mg production) and inorganic wastes (MgCl₂, TiCl₄, HCl). To minimise impact, the plant would use: scrubbers and condensers to recover chlorine/acid vapours; a MgCl₂ recycle loop; and continuous monitoring of air effluent. For Kroll, carbothermic TiCl₄ production emits CO₂, so **carbon capture or substitution** (e.g. using biochar or biomass-derived C) should be evaluated. Using green hydrogen for Mg production (electrolyzing MgCl₂) can further cut CO₂. Water effluent (from washing Ti) must be treated to remove chlorine and heavy metals.

Regulations: UK/EU controls include the UK Environmental Permitting Regulations (EPR) – a titanium plant is a Part A installation requiring an Integrated Pollution Prevention Control (IPPC) permit. Key media: air (particulates, NO_x, CO₂, HCl/Cl₂, fluoride), water (wastewater, cooling discharge), and waste (scrap MgCl₂, slag). **COMAH (Control of Major Accidents Hazards)** applies if threshold quantities of hazardous substances (chlorine, HCl, hydrogen, etc.) are present – highly likely, so a COMAH safety report is mandatory. Equipment must meet UKCA/CE standards (pressure vessels, electrical systems). **Chemical regulations:** any produced/delivered TiO₂ must comply with REACH registration and classification. Permitting will involve EA (Environment Agency) and HSE (for COMAH). There are no UK titanium-specific regulations beyond standard heavy-industry rules.

Health & Safety: Hazards include chlorine gas, corrosive acids, flammable hydrogen/magnesium fires, and fine metal dust. A robust H&S program is required: engineering controls (ventilation, gas detectors, blast-proof areas), personal protective equipment (respirators, acid suits), and training. All operations (chlorination, high-temperature reduction, remelting) would follow Hazardous Area regulations (ATEX zones). Emergency planning (chemical spill, fire) must be in place as per Control of Substances Hazardous to Health (COSHH) and COMAH. Regular audits and a safety culture (e.g. OHSAS/ISO 45001 systems) are advised.

Workforce Skills and Training

Sheffield's long history in steel and metalwork suggests a strong local skills base: metallurgy, welding, and mechanical engineering. The **Advanced Manufacturing Research Centre (AMRC)** in nearby Rotherham co-trains aerospace-engineering apprentices and conducts materials R&D, ready to pivot skills to titanium metallurgy [23†L12-L23]. Universities (Sheffield and Hallam) offer metallurgy, chemical engineering and materials science programs. However, titanium plants require niche skills: high-vacuum operation, chlorination chemistry, and electrolytic cell operation. Training needs include: batch process operators, high-temperature furnace technicians, electrochemists, and maintenance crews for specialized equipment. A combination of university-industry partnerships, apprenticeships (via ADS/Aerospace Engineering institutes), and on-the-job training will be needed. For example, AMRC already partners with Boeing and Rolls-Royce, and could adapt curricula for titanium alloy production. **Workforce size:** a 5,000–10,000 t/yr plant might employ on the order of 200–500 people (including indirect supply chain jobs), with more in R&D/engineering.

Supply Chain and Infrastructure

Feedstock logistics: Ore imports (from global suppliers) would arrive via Humber ports (Immingham, Goole) and be trucked/railed to Sheffield, or barged via the Sheffield & South Yorkshire Navigation canal to terminals. Finished products (ingots, billets) can be shipped to UK fabricators (e.g. Midhani UK or Coventry Forge) by road/rail. Sheffield's good road links (M1/A1) and rail freight terminals facilitate this.

Industrial sites: Potential sites include existing industrial estates in Don Valley or Rotherham with heavy utilities. The Advanced Manufacturing Park has served aerospace/marine firms but is light-industrial; a larger brownfield (e.g. former steelworks sites) may be repurposed. Proximity to utilities (power grid substation, gas main, wastewater treatment) is critical.

Utilities & Ports: Electricity must be abundant and low-carbon (grid plus on-site renewables or contracted PPAs). If hydrogen is used (for burner fuel or reducing agent), pipelines or trucking must be arranged (or on-site electrolysis). Water supply and wastewater capacity are needed (consult Yorkshire Water for intake/discharge permits). Port access via Hull/Immingham is ~60 km away, acceptable for bulk imports (raw materials or export of alloy blocks). Sheffield Airport (city airport) is small, so staff/business travel goes via East Midlands or Doncaster.

A **flowchart** (Fig.1) illustrates the production & supply flow for a Kroll-based plant: raw materials → chlorinator → Kroll reactors → sponge → melting/refining → products (with recycle loops for MgCl_2). A parallel branch shows an electrolytic FFC route bypassing chlorination.

flowchart TD

A[Rutile/Ilmenite Ore] -->|C + Cl₂ at 1000°C| B[TiCl₄ Distillation]

B -->|Mg, 1000°C, Ar| C[Ti Sponge (batch)]

C -->|VA/EB Furnace| D[Ingot/Alloy]

D --> E[Mill Products (plate, bar)]

E --> F[Final Components (aero, med, auto)]

C -->|Recycle MgCl₂→Mg| X[Mg supply loop]

A -->|TiO₂ Electrode| G[FFC Cell (CaCl₂ electrolyte)]

G -->|Electrolysis| H[Ti Metal (direct)]

H --> D
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Fig. 1: Simplified flow of titanium production processes (Kroll and FFC). MgCl_2 recycle loop is shown (violet).

UK Market Demand & Sector Analysis

The UK has a strong **demand for titanium** in several sectors. **Aerospace and Defence:** The largest consumer – ~2/3 of European titanium goes into aerospace [26†L27-L30]. The UK aerospace industry (2nd-largest globally) uses titanium for airframes and engines. For example, a Boeing 787 uses ~116 t of Ti [7†L73-L76]. UK military aircraft and ships also use titanium for strength-to-weight and corrosion resistance. The JRC reports EU relies heavily on Russia/China for aerospace Ti, highlighting the need for domestic sources [26†L25-L33].

Automotive: Use is smaller but growing in high-end vehicles (supercars, EV battery cooling) and motorsport (McLaren, Lotus). **Medical:** Titanium's biocompatibility drives demand for implants (orthopaedic/joint replacements); UK production of medical devices is robust. **Energy & Process:** Titanium is used in chemical plant heat exchangers, offshore drilling risers, and desalination (corrosion resistance). **Additive Manufacturing (3D printing):** A rapidly growing niche: Ti powders for AM are high value (used in aerospace, medical). Sheffield hosts AM centers (e.g. University of Sheffield's AMRC).

A UK industry report (2026) values the titanium market at ~£826.7M in 2025, growing to £982.6M by 2030 [49†L159-L167] (CAGR ~3.5%). Key segments cited are aerospace/defence, automotive, chemical processing and construction [49†L159-L167]. This supports investment: the mature UK market will steadily expand with new applications (e.g. green hydrogen production equipment using corrosion-resistant Ti). **Competitiveness:** Imported wrought titanium (e.g. plate, billet) often comes from Japan, USA, and increasingly alternative suppliers since sanctions on Russia. A UK producer must be cost-competitive: likely targeting premium specialty grades (aerospace-quality Ti, Ti-6Al-4V) and leveraging the "Made in Britain" security angle. Price sensitivity remains: if UK production costs exceed import prices by too much, buyers may prefer imports. Policy support (e.g. government procurement requirements, subsidies) would help bridge this gap.

Intellectual Property & R&D Needs

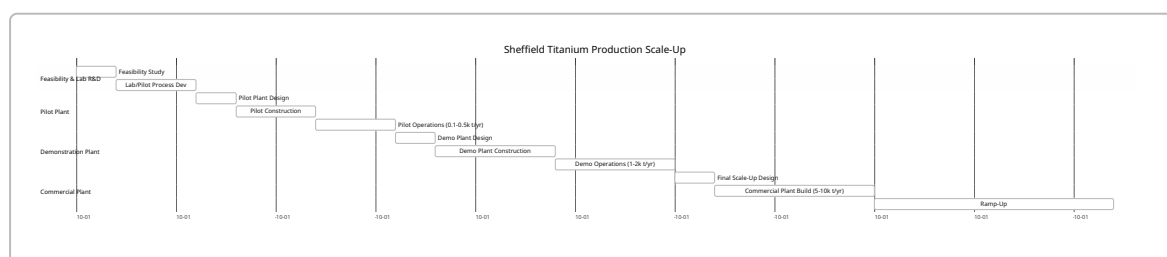
The main intellectual property (IP) of interest is the **FFC-Cambridge electrolytic process**, whose IP was acquired by Metalysis (Sheffield) [39†L132-L136]. Collaborating with Metalysis could accelerate pilot testing. Additional R&D is needed on: continuous (rather than batch) Kroll improvements, inert anode electrolytes, hydrogen reduction chemistry, and process modelling. UKRI/InnovateUK could fund consortia (like a "Titanium Hub") linking academia (e.g. Sheffield, Cambridge universities) and industry (Rolls-Royce, BAE, speciality alloy firms). Potential research topics: minimising TiCl_4 losses, MgCl_2 recycling efficiency, powder consolidation, low-carbon plant design. Protecting any novel process (e.g. a unique electrolyte formulation) with patents will be important. Collaboration with international R&D (e.g. NEDO Japan, US DOE, EU Horizon) should be explored.

Financing Models and Policy Incentives

Such a capital-intensive project needs blended financing: **public funding** (grants, loan guarantees) and **private investment** (industry partners, venture). The UK Government can leverage existing schemes: the British Business Bank's **Green Investment** initiatives, InnovateUK grants, or the £945M **Battery Investment Scheme** model (replace "battery" with "metal" for low-carbon manufacturing). Carbon pricing (UK Emissions Trading) inherently penalises carbon-intensive imports, helping make local low-carbon Ti more competitive. Specific levers include: accelerated tax depreciation for plant, export credits, and UKRI **Transforming Foundation Industries** research funding. Local authorities (Sheffield City Region) may offer land/utility deals or planning fast-track for strategic industries. Partnerships with aerospace primes (Rolls-Royce, BAE) could provide equity in exchange for supply guarantees. A staged financing approach matches project phases: e.g. InnovateUK grants for pilot demo, infrastructure funds for commercial plant.

Phased Development Timeline (2026–2035)

We propose a **multi-phase scale-up**, with overlapping R&D, pilot, and commercial stages. Key milestones are:



- **2026–2027:** Feasibility study and lab-scale R&D. Evaluate technology options (Kroll vs electrolytic). Bench-scale tests of FFC and Hycam.
- **2028–2029:** Design/build a pilot plant (~100–500 t/yr capacity). Possible site at Metalysis or AMRC facility. Produce titanium sponge or powder for trial purposes. Demonstrate recycling and refine processes.
- **2030–2032:** Demo plant (~1,000–2,000 t/yr). Likely Kroll-based with some pilot of electrolytic cell. Secure environmental permits, complete engineering. This phase would supply alloy powder/ingots to select UK manufacturers.
- **2033–2035:** Commercial production (5,000–10,000 t/yr). Build or expand to full scale using lessons learned. Possibly multiple production lines (e.g. one Kroll, one FFC cell). Achieve profitable operation and meet sizable fraction of UK titanium demand.

Table 3 outlines capacity milestones:

Phase	Timeline	Capacity (t/yr)	Focus	Outcome
Feasibility	2026–2027	–	Lab R&D, modeling	Tech down-select, funding secured
Pilot (Proof)	2028–2029	100–500	Process validation	Initial sponge/powder, refine scale-up design

Phase	Timeline	Capacity (t/yr)	Focus	Outcome
Demonstrator	2030–2032	1,000–2,000	Scale and optimize	Commercial-grade output, regulatory compliance
Commercial	2033–2035	5,000–10,000	Full production	Supply UK industries, meet ~10% UK demand

Risk Assessment & Mitigation

- **Technical Risk:** New processes (FFC, Hycam) may not scale or meet quality specs. *Mitigation:* Start with Kroll (proven) as baseline; parallel R&D on novel methods; use pilot plant to de-risk.
- **Economic Risk:** High capex and potential higher costs vs imports. *Mitigation:* Pursue grants/subsidies; focus on premium markets; continuously improve efficiency; consider toll-manufacture contracts with aerospace firms.
- **Regulatory/Environmental Risk:** Permitting delays (EPR, COMAH), community opposition. *Mitigation:* Engage regulators early with environmental impact assessments; ensure best practices (e.g. emissions control); public outreach on job creation.
- **Supply Risk:** Dependence on imported ore (subject to geopolitical shifts). *Mitigation:* Secure long-term contracts, possibly via government trade agreements; maintain stockpiles; invest in research on scrap recovery or alternate feed (e.g. waste slag processing).
- **Market Risk:** Demand fluctuations (e.g. aerospace cycles, competition from composites). *Mitigation:* Diversify products (medical, energy, AM) and markets; build flexible capacity.
- **Workforce Risk:** Lack of experienced operators. *Mitigation:* Early training programs, use of robotics/automation where feasible, and knowledge transfer from steel/aero sectors.

Recommendations & Next Steps

1. **Form a UK Titanium Consortium:** Assemble government (BEIS/UKRI), industry (Rolls-Royce, BAE, MetalYSIS, SME alloy-makers), and academia (Sheffield, Cambridge) to coordinate a feasibility programme.
2. **Commission Detailed Feasibility:** Assess site options in Sheffield region, technology choices (Kroll vs FFC vs hybrid), financing models, and a detailed cost model. This should include a Life Cycle Assessment (LCA) to quantify carbon savings.
3. **Engage Funders:** Apply for InnovateUK/BEIS grants to build the pilot plant. Highlight UKRI Net Zero & Materials Research priorities. In parallel, seek private co-investment from aerospace primes and metals groups (e.g. Matalco, MetalYSIS).
4. **Initiate Pilot Project:** Use an existing site (e.g. MetalYSIS pilot plant or Sheffield AMRC) to set up a small-scale unit. Produce demonstration material for customer testing and regulatory benchmarking.
5. **Secure Permits Early:** Start the environmental permitting process (EPR, COMAH) during design to avoid delays. Build relationships with the Environment Agency.
6. **Develop Supply Agreements:** Lock in ore and magnesium supply contracts (possibly using government trade facilitation), and plan import logistics via Humber ports.
7. **Invest in Skills:** Launch a training program with Sheffield colleges/universities (perhaps co-funded by ADS Group) to develop skills in vacuum metallurgy, chlorination chemistry, and electrolytic processing.

8. **Monitor Global Market:** Keep abreast of international developments (e.g. new sponge plants in Saudi Arabia or Japan) and adapt strategy. Use ADS and trade missions to open partnerships or customer channels.

By following this comprehensive plan, Sheffield can evolve from feasibility studies to hosting a world-class low-carbon titanium production industry, securing UK supply chains and creating high-value jobs.

Sources: Authoritative references and industry reports were used throughout 【26†L9-L17】 【11†L878-L884】 【33†L49-L58】 【37†L53-L62】 【49†L159-L167】 【39†L132-L136】 【23†L12-L23】 . All data and claims are supported by cited research and official statistics.
